50.007 Machine Learning

Hidden Markov Model

(Continue…)

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Learning Objectives

You should be able to know:

- What is decoding for the HMM?
- Why do we need the Viterbi algorithm for decoding?
- How to perform decoding using the Viterbi algorithm?
- What is the space and time complexity of the Viterbi algorithm?

HMM - Supervised Learning (Recap)

We estimate the parameters of the HMM by performing Maximum Likelihood Estimation.

$$p(\mathbf{x}, \mathbf{y}) = \underbrace{\prod_{j=0}^{n} a_{y_j, y_{j+1}}}_{\text{Transition probabilities}} \times \underbrace{\prod_{j=1}^{n} b_{y_j}(x_j)}_{\text{Emission probabilities}}$$

By setting the derivative of log-likelihood *w.r.t.* each individual parameter to zero, we can get a close-form estimate for each parameter:

Emission matrix

transition matrix

$$a_{u,v} = \frac{\text{count}(u,v)}{\text{count}(u)}$$

Number of times we see a transition from u to v

Number of times we see the state u in the training set

$$b_u(o) = \frac{\text{count}(u \rightarrow o)}{\text{count}(u)}$$

Number of times we see observation o generated from u

Number of times we see the state u in the training set

HMM (Example)

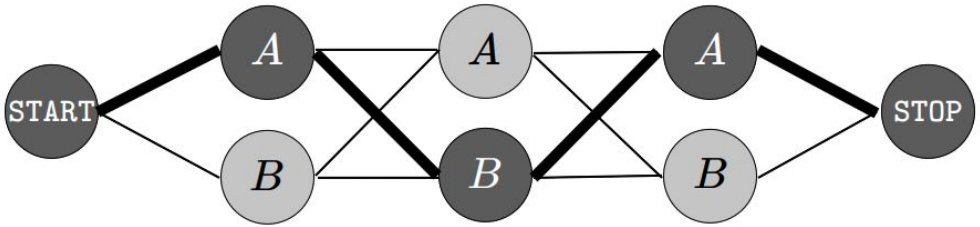
$a_{u,v}$

$u \backslash v$	A	B	STOP
START	1.0	0.0	0.0
A	0.5	0.5	0.0
B	0.0	0.8	0.2

$b_u(o)$

$u \backslash o$	*"the"*	*"dog"*
A	0.9	0.1
B	0.1	0.9

$$(\mathbf{x}, \mathbf{y}) = \text{the}/A, \text{dog}/B, \text{the}/A$$



$$a_{\text{START},A} \times b_A(\text{“the”}) \times a_{A,B} \times b_B(\text{“dog”}) \times a_{B,A} \times b_A(\text{“the”}) \times a_{A,\text{STOP}} = 0$$

HMM (Example)

$a_{u,v}$

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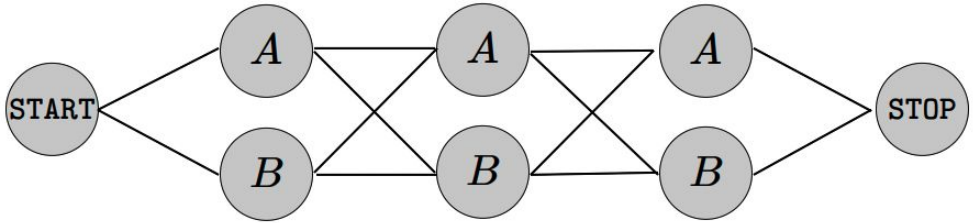
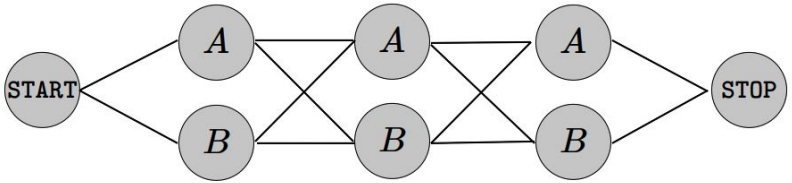


Which label sequence y is the most probable given the word sequence x?

Decoding

HMM - Decoding

$$(\mathbf{x}, \mathbf{y}) = \text{the}/B, \text{dog}/B, \text{the}/B$$



- Given the HMM (with estimated parameters) and an observation sequence **x**, find the most probable sequence of hidden states **y^*** for the given observation sequence **x**.

$$\mathbf{y}^* = \arg\max_{\mathbf{y}} p(\mathbf{y}|\mathbf{x})$$

↓

Return the parameters of y which gives the max. $P(y|x)$

HMM - Decoding

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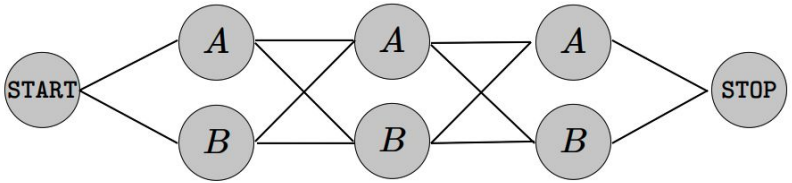

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$$\mathbf{y}^* = \arg\max_{\mathbf{y}} p(\mathbf{y}|\mathbf{x})$$

$$= \arg\max_{\mathbf{y}} p(\mathbf{x}, \mathbf{y})/p(\mathbf{x})$$

Bayes Theorem

will be constant ∵ it is the P of observing "the", "Dog", "the"

⇒ essentially able to drop

HMM - Decoding

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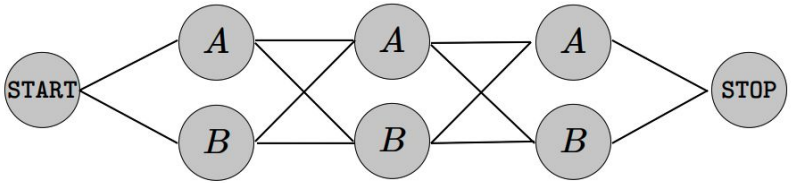

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HMM - Decoding

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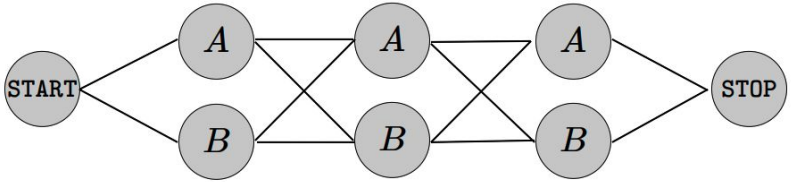

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Brute forcing

We can try one **y** at a time, and see which gives the highest score!

HMM - Decoding

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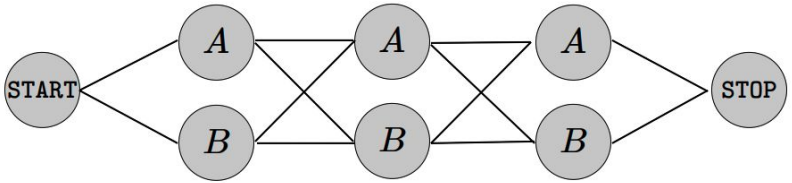

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HMM - Decoding

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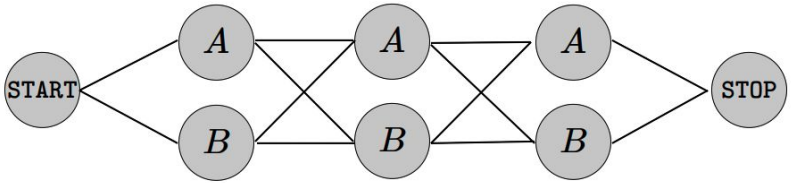


$$\mathbf{y}^* = \arg\max_{\mathbf{y}} p(\mathbf{x}, \mathbf{y})$$

Number of words in the sentence → # of words

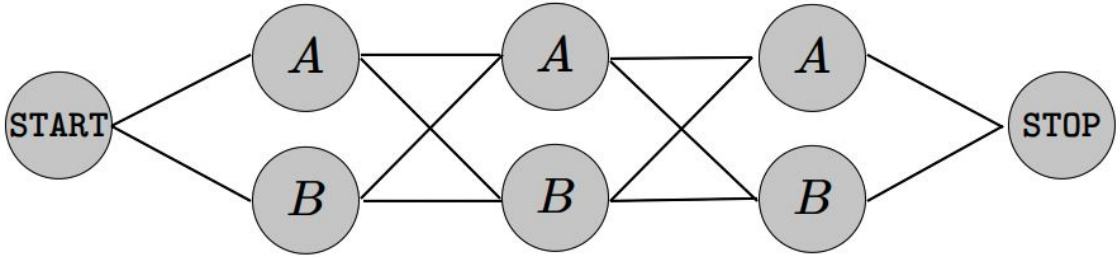
There are $O(|\mathcal{T}|^n)$ possible $\mathbf{y}$'s!

Number of possible tags at each word/position

2^3 # of hidden state

Finding Highest Scoring Path

- We aim to develop a more efficient approach for finding the most probable label sequence, or, the the highest scoring path connecting START and STOP.



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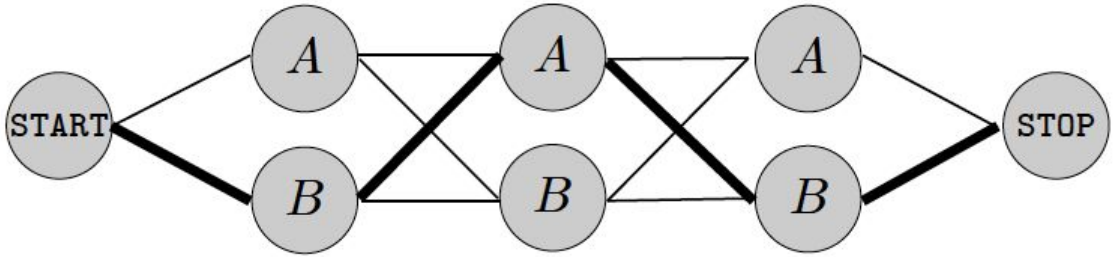

Let's assume this is the highest scoring path.

Finding Highest Scoring Path

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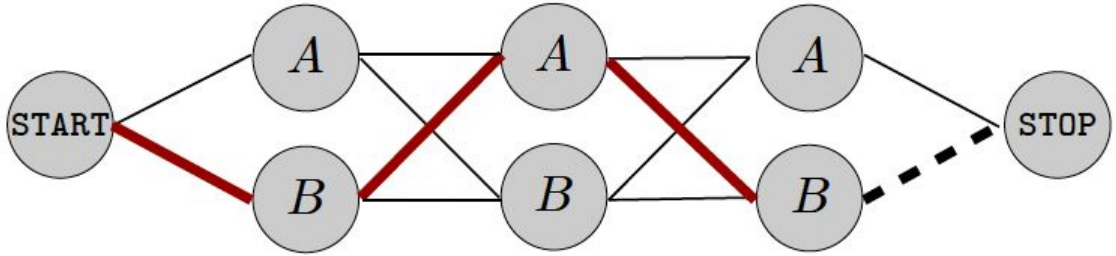

Let's assume this is the highest scoring path.

Then, what can we say about this partial path?

(What types of properties do we know for this path?)

Finding Highest Scoring Path

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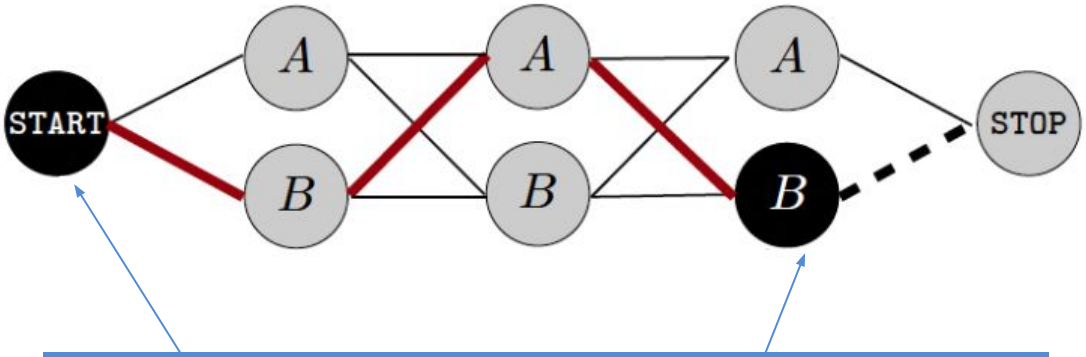

Finding Highest Scoring Path

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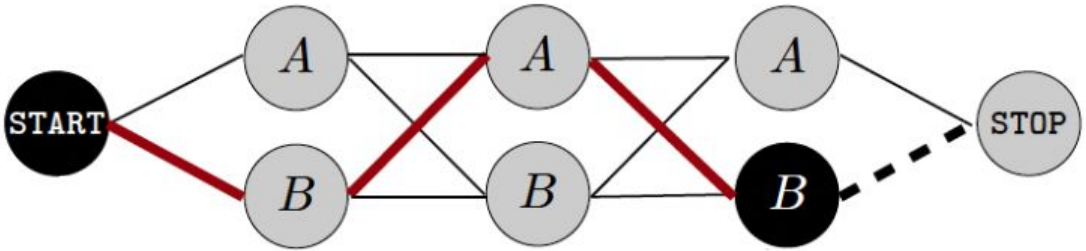
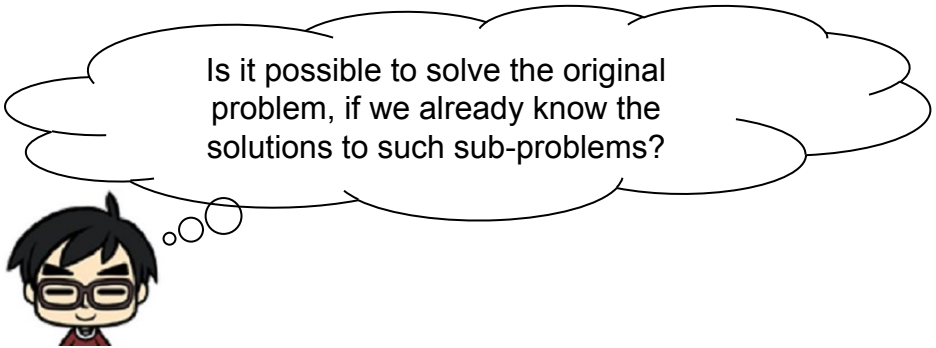
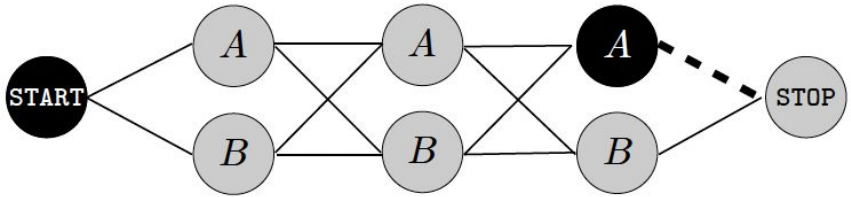




Finding Highest Scoring Path

Case A

The second last node in the highest scoring path is A.



Find the highest scoring path from START to A at position n

A single edge from node A to STOP

Finding Highest Scoring Path

Case A

The second last node in the highest scoring path is A.

Find the highest scoring path from START to A at position n

A single edge from node A to STOP

Finding Highest Scoring Path

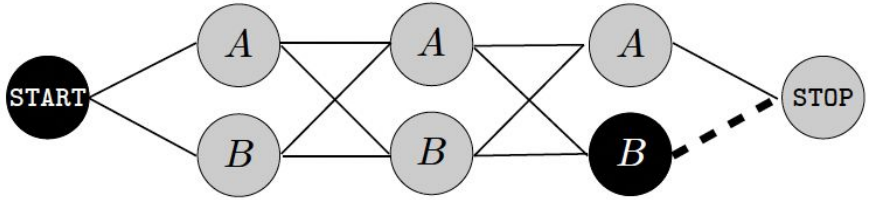
Case A
The second last node in the highest scoring path is A.

Find the highest scoring path from START to A at position n

+

A single edge from node A to STOP

Case B
The second last node in the highest scoring path is B.



Find the highest scoring path from START to B at position n

+

A single edge from node B to STOP

Finding Highest Scoring Path

Case A
The second last node in the highest scoring path is A.

Find the highest scoring path from START to A at position n

A single edge from node A to STOP

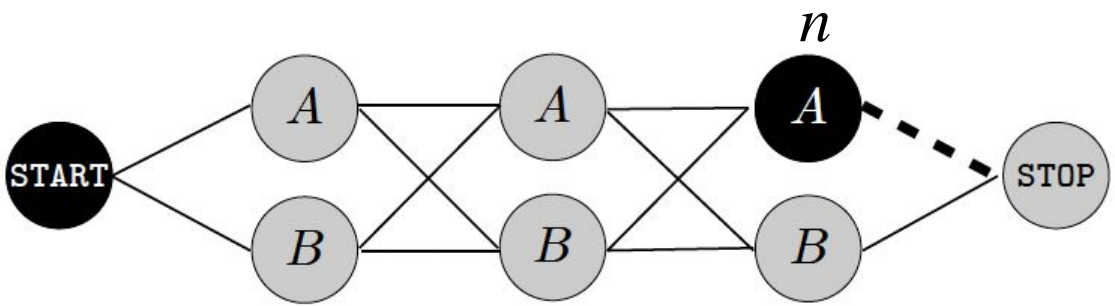
Case B
The second last node in the highest scoring path is B.

Find the highest scoring path from START to B at position n

A single edge from node B to STOP

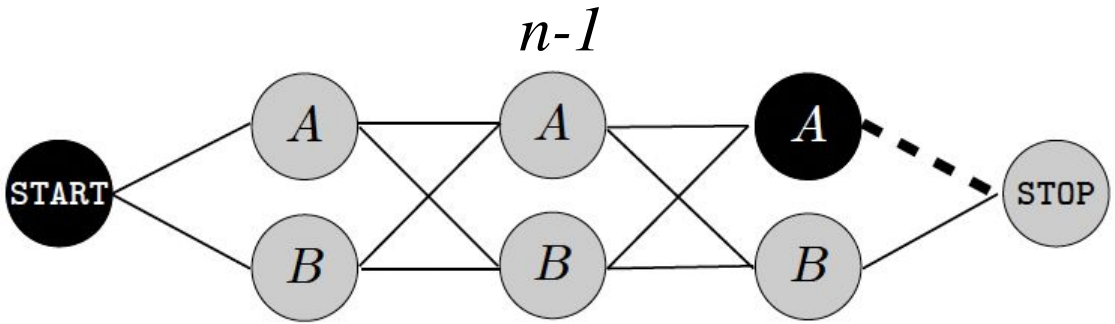
The highest scoring path from START to STOP: PATH 1 or PATH 2?

Finding Highest Scoring Path



How do you find the highest scoring path from START to node A at position n?

Finding Highest Scoring Path



How do you find the highest scoring path from START to node A at position n?

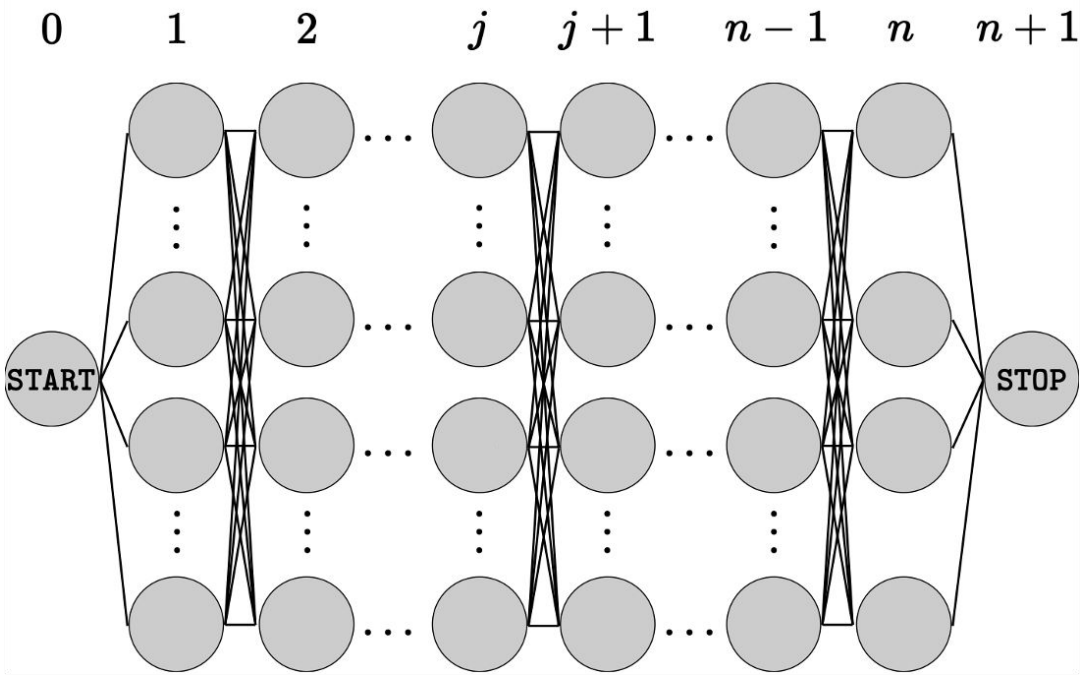
We shall again rely on the partial paths from START to the two nodes at position $(n-1)$

Dynamic Programming

Question


Now, can we develop an efficient algorithm based on such observations?

Viterbi Algorithm



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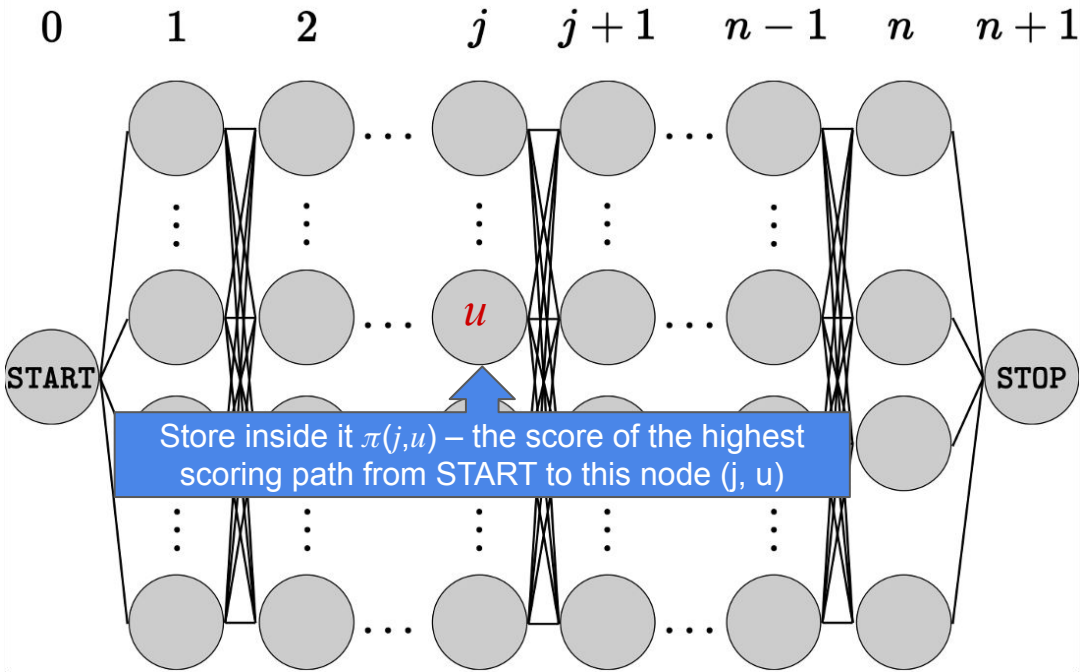
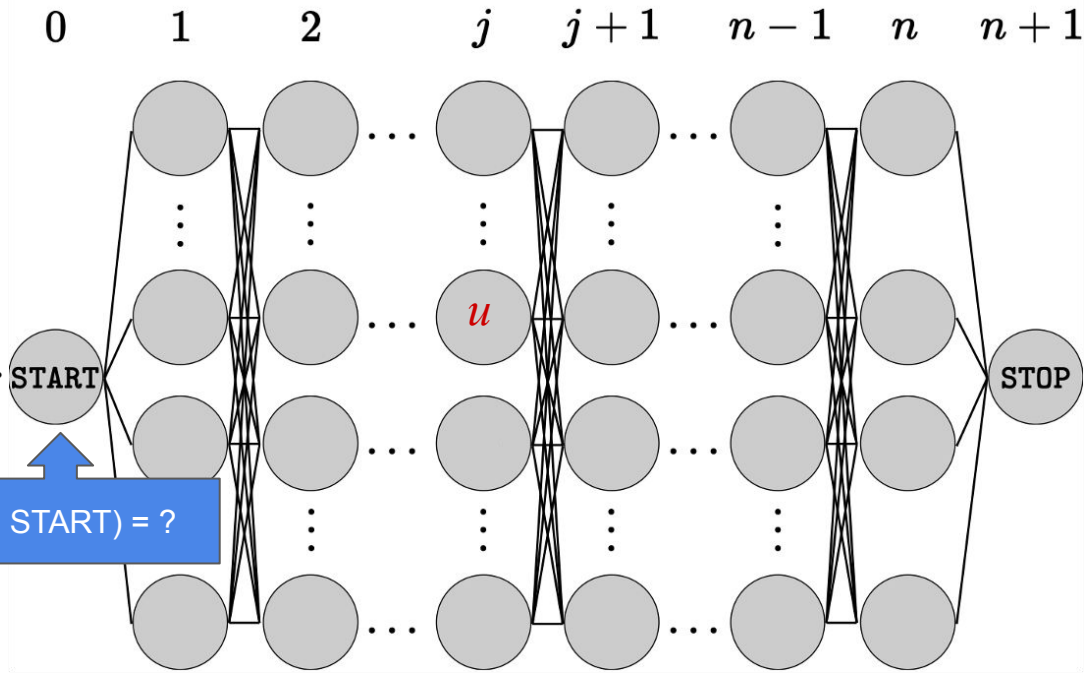


$$\pi(j,u) = \max_{y_1,\ldots,y_{j-1}} \{p(x_1,\ldots,x_j, y_0 = \text{START}, y_1,\ldots,y_{j-1}, y_j = u)\}$$

storing the max. value instead of argmax.

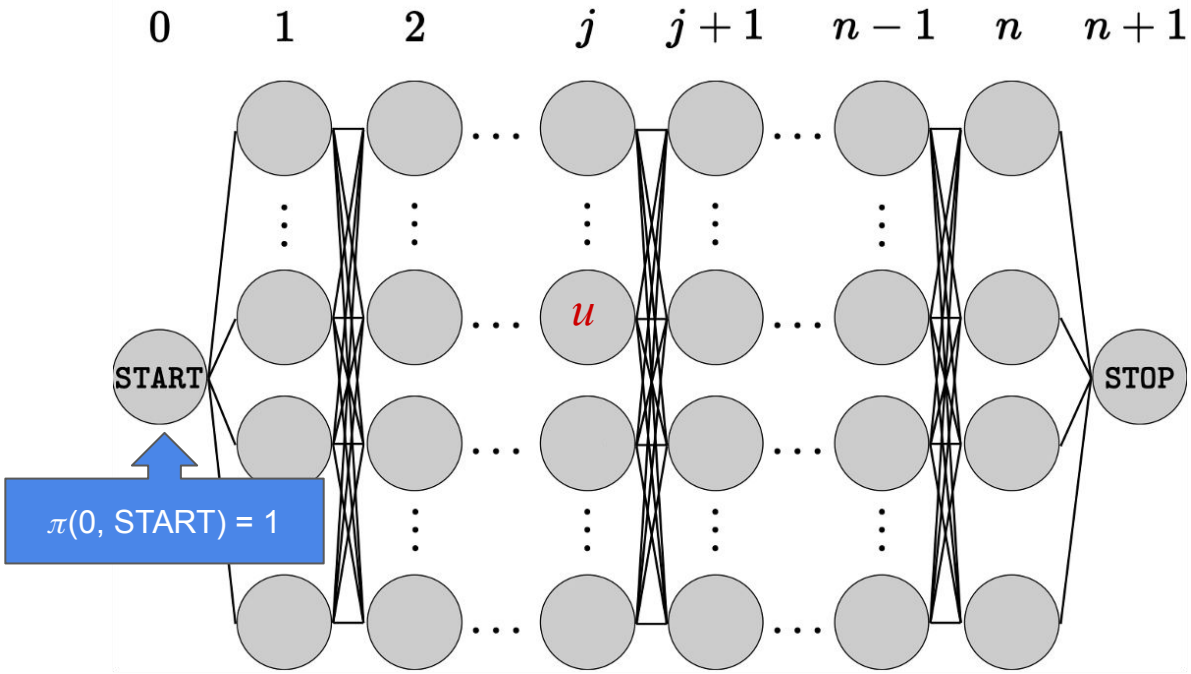
Viterbi Algorithm



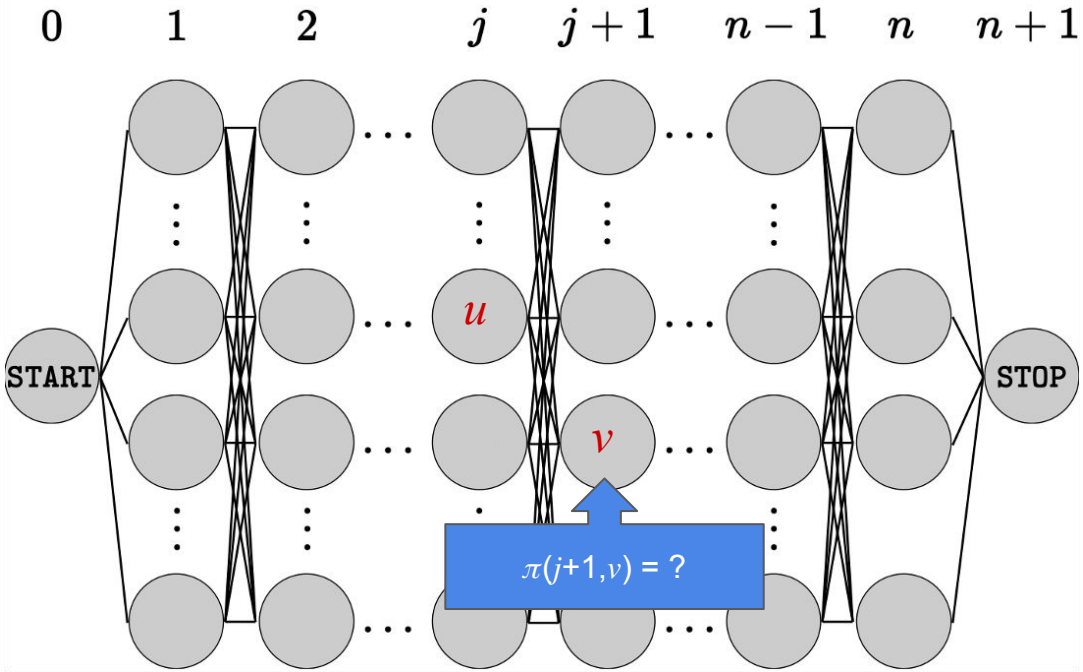
Initialized to 1
∴ it multiplies with all other probabilities.

$$\pi(0, \text{START}) = p(y_0 = \text{START})$$

Viterbi Algorithm

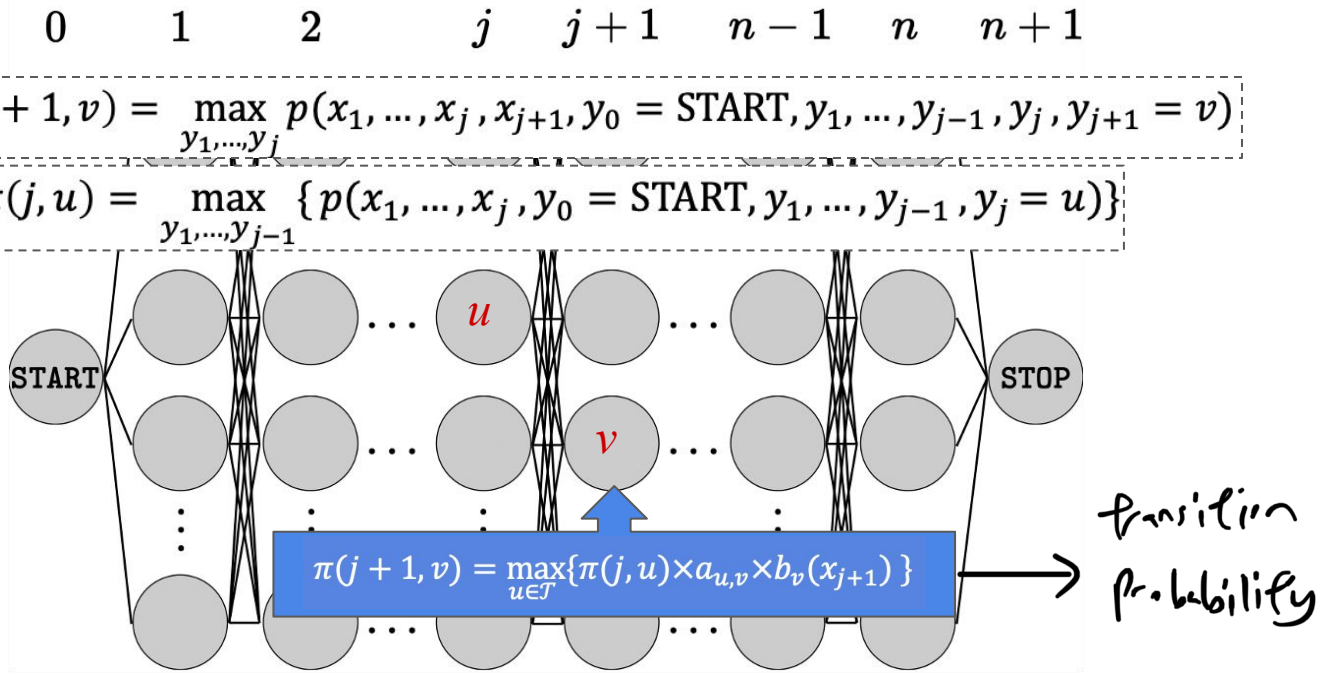


Viterbi Algorithm

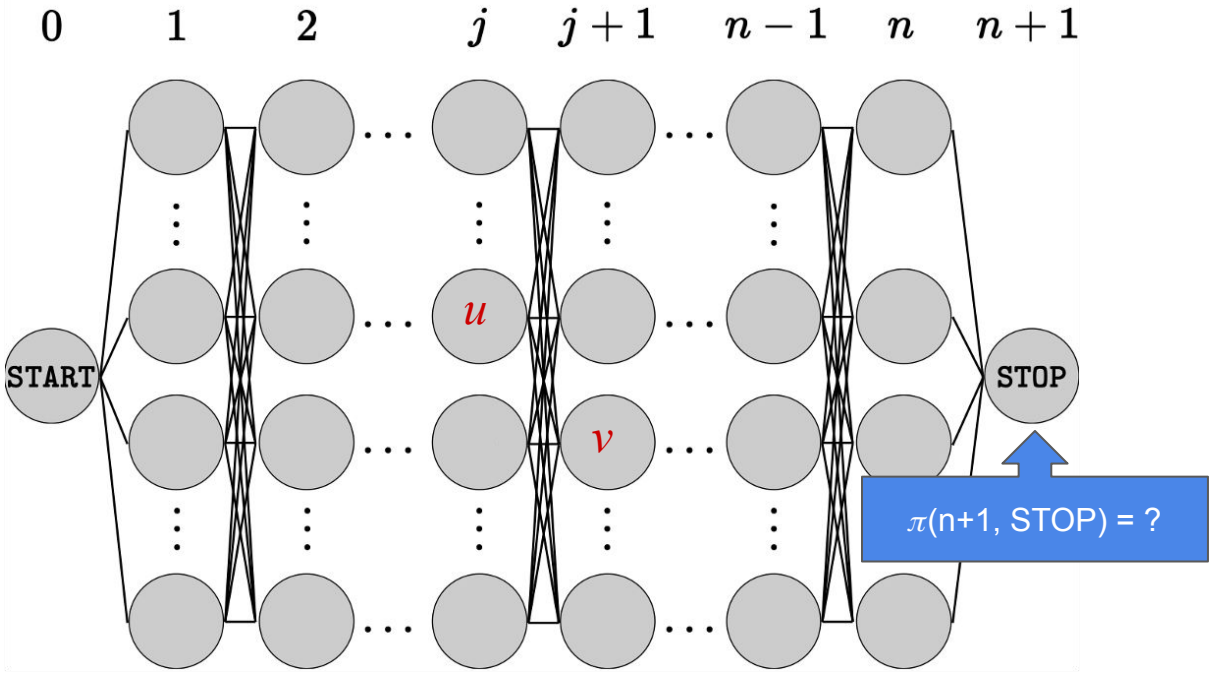


$$\pi(j+1, v) = \max_{y_1, \ldots, y_j} p(x_1, \ldots, x_j, x_{j+1}, y_0 = \text{START}, y_1, \ldots, y_{j-1}, y_j, y_{j+1} = v)$$

Viterbi Algorithm



Viterbi Algorithm



Viterbi Algorithm

*similar to Bellman-Ford

- Initialization

$$\pi(0,u) = \begin{cases} 1 & if\ u = \text{START} \\ 0 & otherwise \end{cases}$$

- For $j = 0, \ldots, n-1$:
  for each $v \in \mathcal{T}$:

$$\pi(j+1,v) = \max_{u \in \mathcal{T}} \{\pi(j,u) \times a_{u,v} \times b_v(x_{j+1})\}$$

- Final Step

$$\pi(n+1, \text{STOP}) = \max_{u \in \mathcal{T}} \{\pi(n,u) \times a_{u,\text{STOP}}\}$$

Viterbi Algorithm

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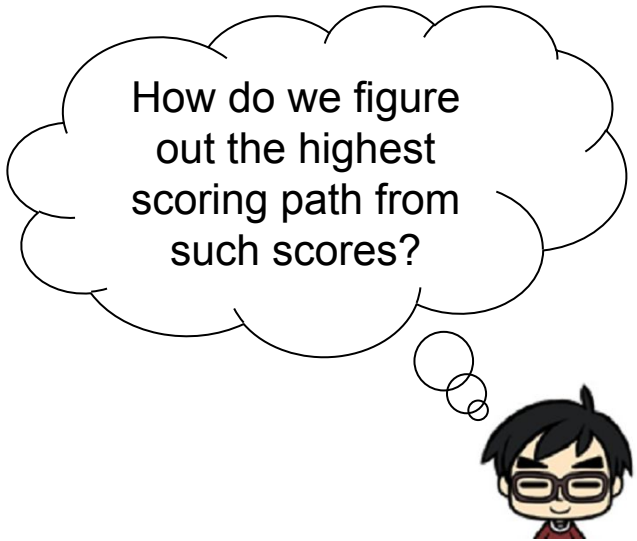


Viterbi Algorithm - Backtracking

Now, having values $\pi(j, u)$, we reconstruct the most likely sequence of tags $\mathbf{y}^* = \arg\max_{\mathbf{y}} p(\mathbf{x}, \mathbf{y})$ via **backtracking**:

- The best value for the last hidden random variable:

$$y_n^* = \underset{u \in \mathcal{T}}{\text{argmax}} \{\pi(n, u) \times a_{u,\text{STOP}}\}$$

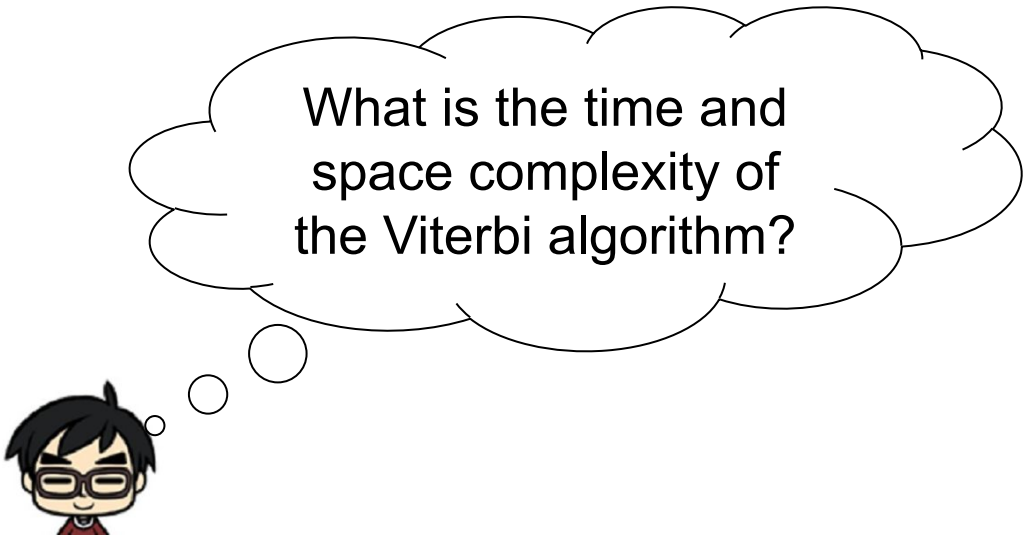
- For $j = n - 1, \ldots, 1$:

$$y_j^* = \underset{u \in \mathcal{T}}{\text{argmax}} \{\pi(j, u) \times a_{u,y_{j+1}^*} \times b_{y_{j+1}^*}(x_{j+1})\}$$

$$y_j^* = \underset{u \in \mathcal{T}}{\text{argmax}} \{\pi(j, u) \times a_{u,y_{j+1}^*}\}$$

Drop $b_{y_{j+1}^*}(x_{j+1})$ since it is irrelevant to u at j-th position.

Question


What is the time and space complexity of the Viterbi algorithm?

Viterbi Algorithm - Time & Space Complexity

- Initialization

$$\pi(0,u) = \begin{cases} 1 & if\ u = \text{START} \\ 0 & otherwise \end{cases}$$

- For $j = 0, \ldots, n-1$: n

  for each $v \in \mathcal{T}$: $|\mathcal{T}|$

  $$\pi(j+1,v) = \max_{u \in \mathcal{T}} \{\pi(j,u) \times a_{u,v} \times b_v(x_{j+1})\}$$ $|\mathcal{T}|$

  $n|\mathcal{T}|$

- Final Step

$$\pi(n+1, \text{STOP}) = \max_{u \in \mathcal{T}} \{\pi(n,u) \times a_{u,\text{STOP}}\}$$

Time Complexity: $O(n|\mathcal{T}|^2)$

Space Complexity: $O(n|\mathcal{T}|)$

What you should know?

- What is decoding for the HMM?
- Why do we need the Viterbi algorithm for decoding?
- How to perform decoding using the Viterbi algorithm?
- What is the space and time complexity of the Viterbi algorithm?

Suggestion: If you're not very clear or want to learn more, please do read: "C. Bishop: Pattern Recognition and Machine Learning. Springer, 2006" (recommended text book). It will help you to understand the concept.

Acknowledgement

- *The following material have been referenced or partially used:*
  - *Lecture notes from Prof. Lu Wei*
  - *Lecture notes from Prof. Zhao Na*
  - *National University of Singapore – Pattern Recognition Module (EE5907R)*
  - *National University of Singapore – Neural Networks Module (EE5904R)*
  - *University of Edinburgh, United Kingdom – Machine Learning And Pattern Recognition (MLPR, INFR11130)*
  - *Stanford University – Machine Learning (CS229)*